Technical requirements of the Android project

The project is chosen by the students and must be supervised by the lab instructor in order to avoid too large / too small project sizes and unachievable program features. Students must work in groups (2 to 4 students).

Technical requirements

- Android Studio IDE (**all team members must use the same version!**)
- Android SDK (**all team members must use the same version!**)
- Java programming language
- git version control (use git bash / sourcetree / other git tools)
    - students must commit frequently
    - upload every commits to the server, only the commits located on the server count (merge commit type does not count)
- store your project in GitHub or Gitlab online
    - invite the lab instructor for the Collaborator/Reporter role
- readme.md must contain a brief summary, the goal of the project and the app features described as bullet points
- optimize your app resolution for only one specific resolution
- use the portrait display mode in your app (landscape mode is used only in justified cases, eg.: game app)

Implementation

- App consists of one Activity and multiple Fragments
    - using multiple Activities are prohibited
- Use **ConstraintLayout** as a layout management
- Create list and detail pages in your app
    - Depending on the project, create a page for “add new element”, updating an existing element and a feature to delete items
    - Depending on the project search and ordering functionality
- Use **RecyclerView** for listing items
- Tab, Action Bar or **Navigation Drawer** are used for navigation
- In case of a local database, use Architecture Components Room
- Integration with a REST API (or any kind of server-side component) is implemented with Retrofit of Volley packages
- Use Glide package to display large resolution images

Example of 3 students project

The Movie Database (TMDb) (https://developers.themoviedb.org/3) webpage provides detailed information about today's movies and series. The goal of the project is to display data from the Movie Database. In order to do this, an integration is needed to the REST API of themoviedb.org. The app contains an offline (local) database to store your favourite movies or series.

Features to be implemented

- Display movies in a list
- Implement navigation from the movie list page to the details
  - Display the poster image of the movie
  - Add the current movie to favourites
- Implement a search functionality (ex.: search by movie title, authors, release year, etc.) and display the searching results as a list
- Display series in a list
- Implement navigation from the series list page to the details
  - Display the poster image of the series
  - Add the current series to favourites
- Display the favourite movies and series in a list
  - Store favourites in an offline/local database

Technical information

- The navigation among three list views (movies, series, favourites) are implemented with Tab Bar/Button Navigation
- List views are implemented with RecyclerView
- REST API endpoint integration is implemented with Retrofit or Volley packages
- Image display is implemented with Glide package
- Local (offline) database is implemented with Architecture Components Room package